

Exploring metal alloys Review paper

Introduction

One of the best methods to learn about a new topic such as metal alloys is to read a review paper.

The aim of this activity is to introduce new aspects of metal alloys by critically reading a scientific review, more specifically

1. Describe new vocabulary specific for the nomenclature of metal alloys
2. Present data and its interpretation of the surface characterization of metal alloys using electron microscopy
3. Summarise applications of metal alloys

Task

You will learn about metal alloys using as example the recently described aluminium metal matrix composites. The information can be found in the review paper by Kumar and Venkatesh (2019) that was uploaded to QMPlus.

Read the paper paying special attention to the different sections. A useful method for reading scientific papers was described by Raff (2018). You can use this as a guide, the document is in QMPlus.

There is also an infographic guide produced by Elsevier that will help you to follow the steps to read and understand a scientific paper (see infographics in QMPlus).

During the process, you should take the opportunity to reflect on the following questions:

Introduction

1. What was the authors' rationale for the study? To aid you in your answer, indicate the broad scale topic that they felt their questions related to.
2. What question or questions are the authors seeking to address?

Methods

1. How does the design of the study address the question(s) posed?
2. What methodological approaches were used to obtain data for this study?
3. What was the data being collected?
4. What statistical tests (if any) did the authors use to assess their data?

Results

1. What were the key results of this study? (Pay particular attention to tables and figures and where they are referenced in the text.)
2. Do the results represent the overall question of the study?

Discussion

1. Did the authors conclude that their research was supported by their findings?
2. Does this agree with your assessment of their results?
3. Be thoughtful about their interpretation of the results. Do they justify their interpretation well? Does it make sense in light of the literature they cited? Why or why not?
4. How do the authors relate their findings to their rationale or broader scale applications?
5. How do they (or would you) suggest following up on their study? In other words, what experiments or studies could be done to build on their work and/or to address unanswered questions?

Write your reflection as you focus your reading through the document. Take good notes by hand or electronically.

Make sure you prepare a glossary of new words. These will help you understand other aspects of your studies and career.

This is a non-assessed formative homework. Completing this work will contribute to your overall development as a professional in Materials and Polymer Science.